Department of Electrical, Electronic, and Computer Engineering

2019EBN111E01 QUESTION PAPER

Module EBN111 Electricity and Electronics

June 2019

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA
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Exam Information:			
Maximum marks:	92	Full marks:	90
Duration of exam:	200 minutes	Open/closed book	Closed
Total number of pages (including this page):		32 + OCR	

IMPORTANT

1. *The examination regulations of the University of Pretoria apply.*
2. ***All questions must be answered on the Optical Character Recognition (OCR) form.***
3. *Carefully read and follow the instructions on the OCR form.*
4. *Answers must include units when applicable!*
5. ***Final answers must be written as real numbers, and not as fractions.***
6. *Round off answers to 3 significant digits when applicable.*

Lecturers	1. D Ramotsoela
	2. X. Ye
Internal Moderator	F. Paluncic

INSTRUCTIONS

Do step 0 in the first 5 minutes of the *session*. (-5 min to 0 min)

Do step 1 in the first 180 minutes of the *exam*. (0 min to 180 min)

Do step 2 within the following 20 minutes. (180 min to 200 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form. Also write the assessment ID and student number on ALL pages of the OCR form.

STEP 1 (0 min to 180 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units when applicable. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (180 min to 200 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2019EBN111E01**
- Use a mechanical pencil with 0.5mm HB lead to complete the OCR form.
- Each cell must only have one character.
- Characters must not touch or cross the cell boundaries.
- **Do not use commas!** Use full stops.

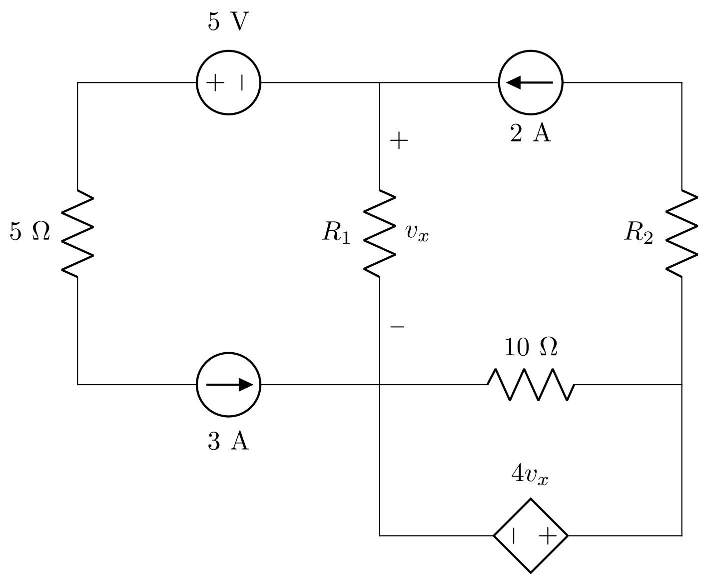
Examples

01	6	8	.	4					°									
02	3	4	0	+	j	2	0		Ω									
03	1	5	.	2	∠	4	2	°	V									
04	1	7							r	a	d	÷	s					

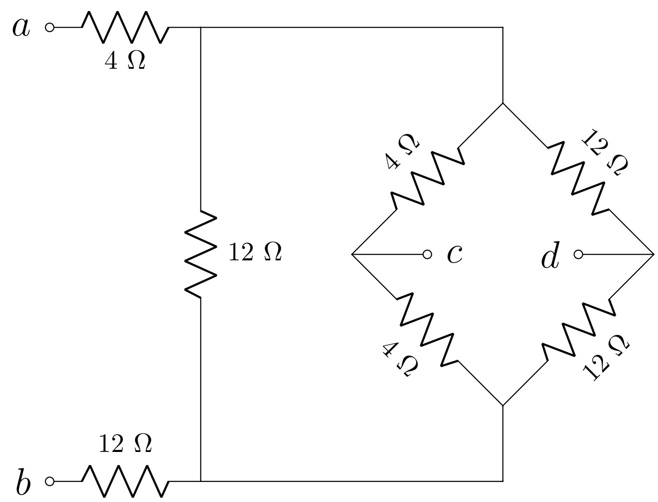
SECTION 1 [46]

Questions 1–2 [4]	
01	A current of 2 A enters the negative terminal of an element between 0.5 s and 1 s. Determine the energy associated with the element in this period if the terminal voltage is $v(t) = 5\left(t - e^{-0.5t}\right)$ V.
02	A 6 kW oven is able to roast 4 chickens in 120 minutes. A restaurant uses this oven to prepare 16 chickens every day. If electricity costs R $3/\mathrm{kWh}$, how much does the restaurant spend on electricity (per month of 30 days)?

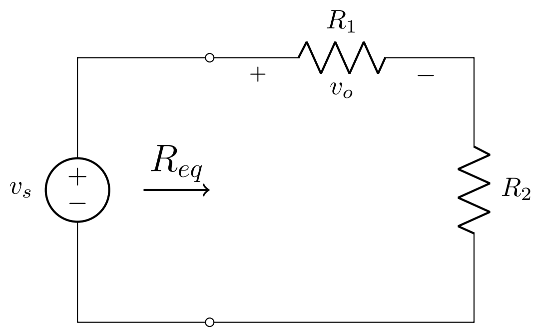
Questions 3–4 [4]	
Assume that $v_x = 5$ V,	
03	Find the power associated with the 3 A current source (P_I).
04	Find the power associated with the dependent voltage source (P_V).



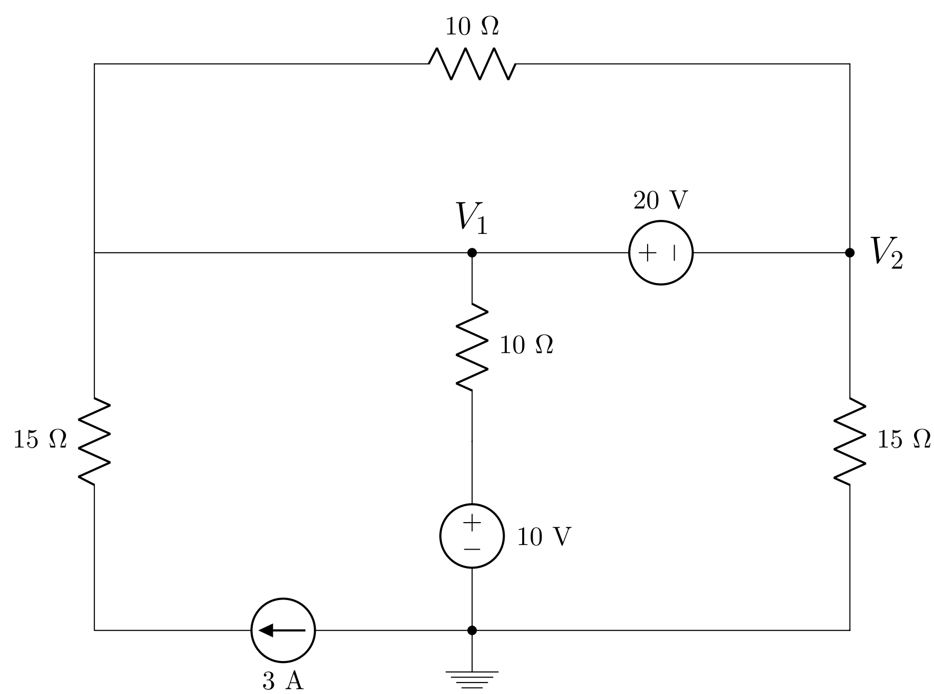
Questions 5–6 [4]	
05	Find the equivalent resistance w.r.t. terminals $a - b$ (R_{ab}).
06	Find the equivalent resistance w.r.t. terminals $c - d$ (R_{cd}).



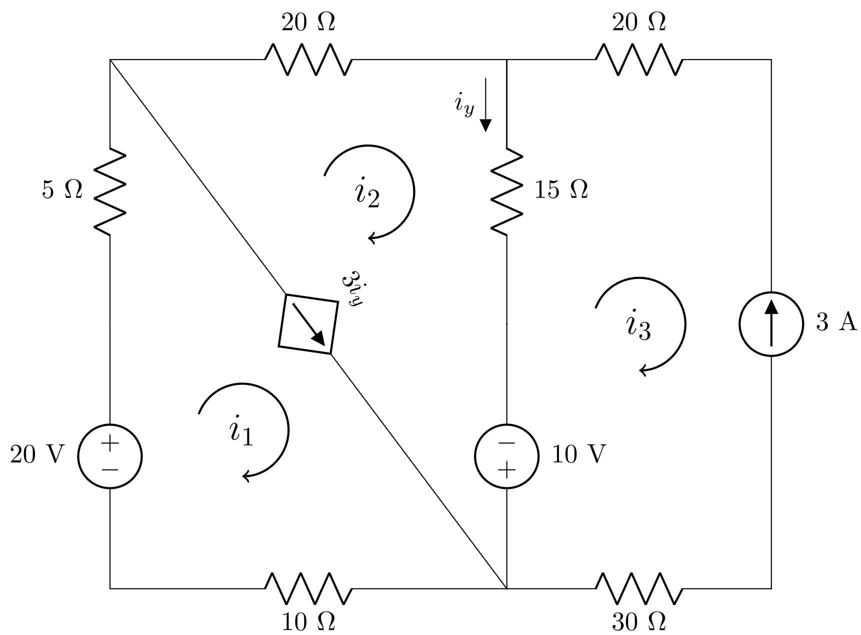
Question 7 [2]	
07	Find R_2 if $\frac{v_o}{v_s} = 0.05$ and $R_{eq} = 20$ kΩ.



Questions 8–11 [4]			
Use nodal analysis to set up two equations of the following form and find the values a, b, c, and d. $$aV_1 + bV_2 = 20$$ $$cV_1 + dV_2 = 120$$			
08	*a*	**09**	*b*
10	*c*	**11**	*d*

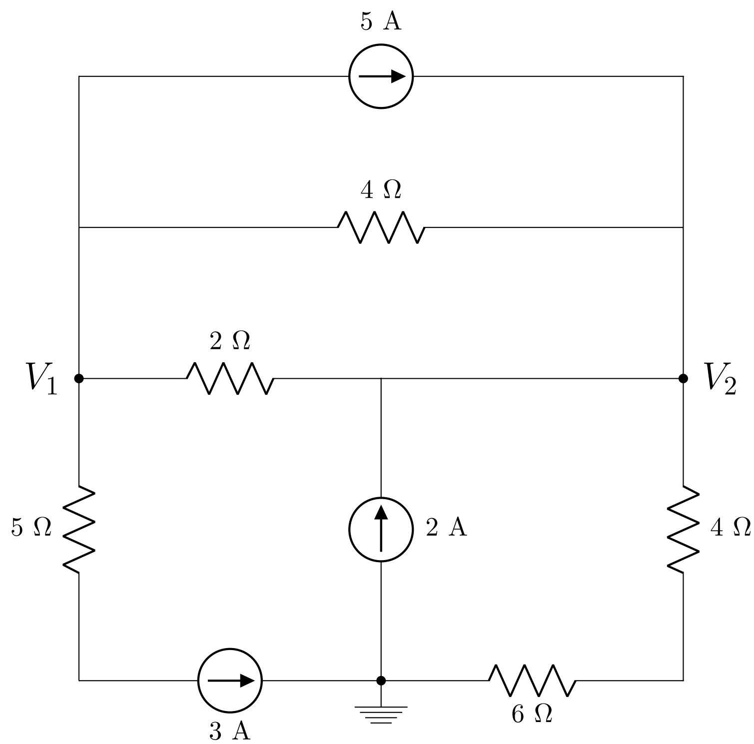


Questions 12–15 [4]			
Use mesh analysis to set up two equations of the following form and find the values e, f, g, and h. $$ei_1 + fi_2 = 9$$ $$15i_1 + gi_2 = h$$			
12	*e*	**13**	*f*
14	*g*	**15**	*h*

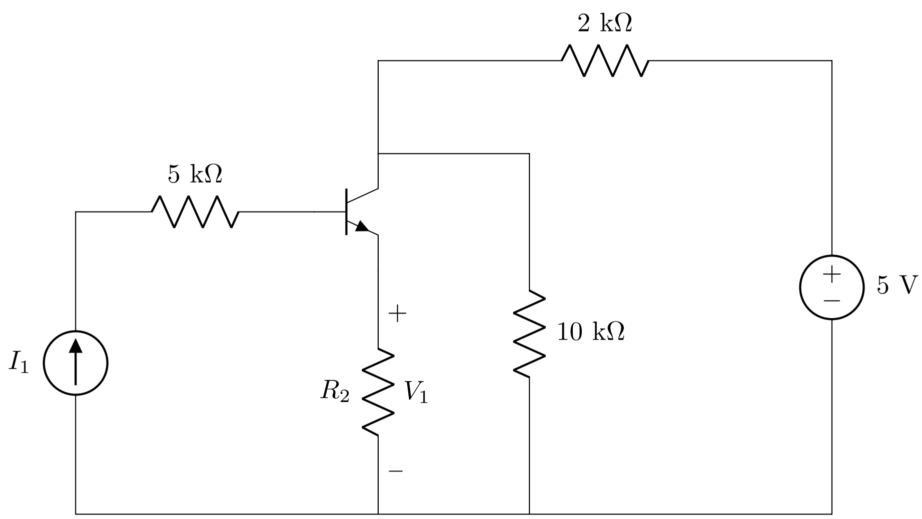


Questions 16–17 [2]			
Use inspection to set up a matrix equation of the following form and find the values i and j.			
16	i	**17**	j

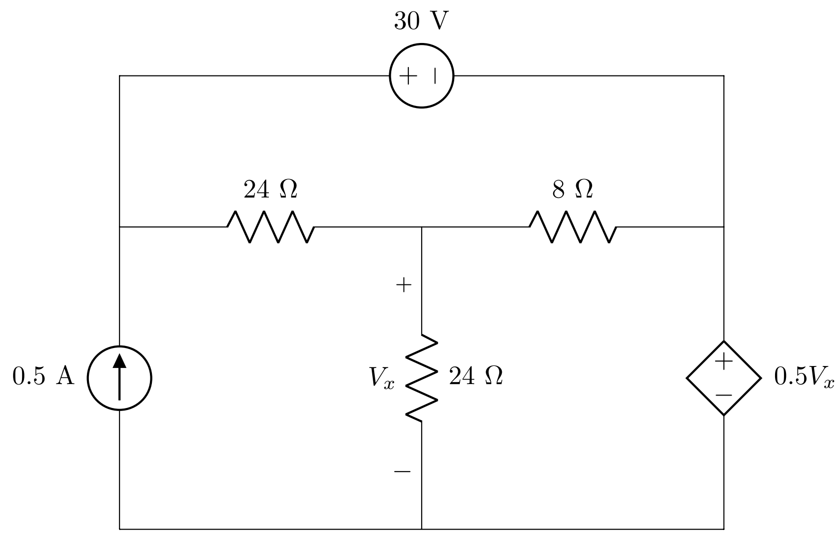
$$\begin{bmatrix} 0.75 & -0.75 \\ -0.75 & j \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} i \\ 7 \end{bmatrix}$$



Questions 18–19 [4]	
In the circuit below, $\beta = 99$, and $V_{BE} = 0.7$ V,	
18	Find I_1 if $V_1 = 3$ V and $R_2 = 2$ kΩ.
19	Find V_{CE} if $R_2 = 0\ \Omega$ and $I_1 = 12\ \mu$A.

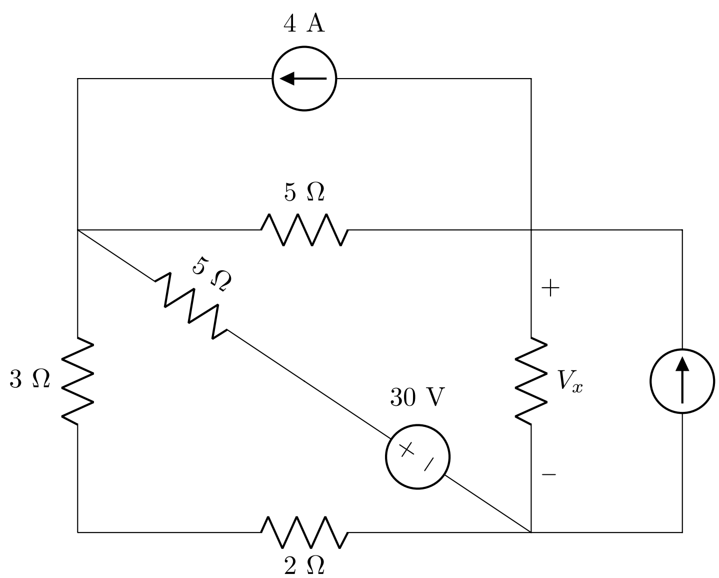


Questions 20–21 [4]	
20	Find V_x^V, the contribution of the independent voltage source to V_x.
21	Find V_x^I, the contribution of the current source to V_x.

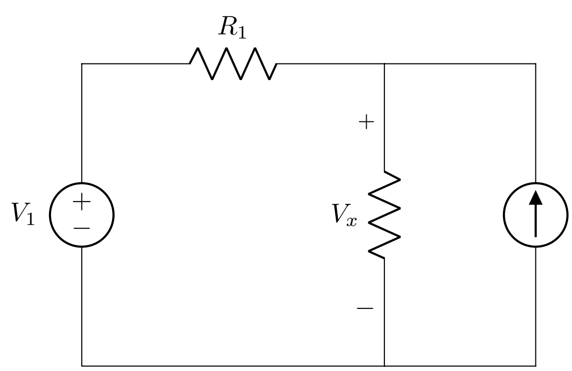


Questions 22–23 [4]	
Use source transformation to determine V_1 and R_1 in ***circuit B*** such that it is equivalent to ***circuit A***.	
22	V_1
23	R_1

Circuit A:



Circuit B:



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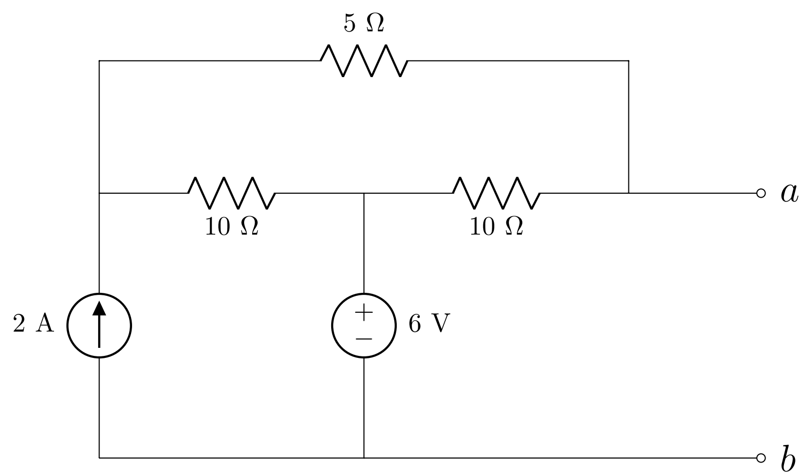
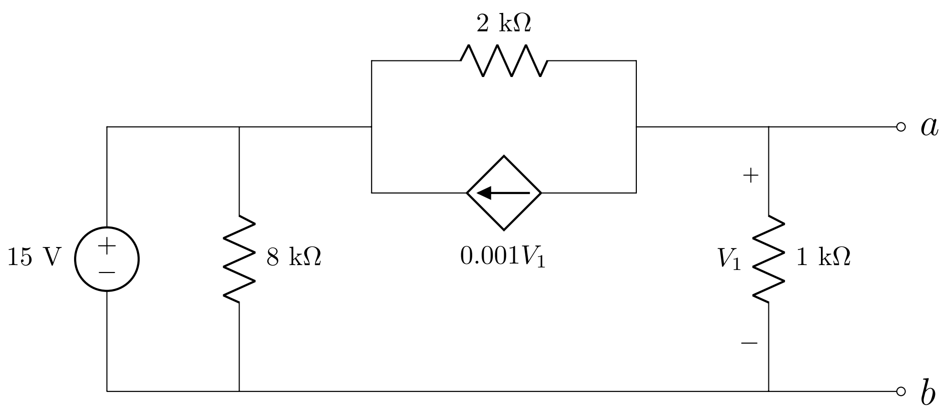
Additional space for Questions 22–23.

Questions 24–25 [4]	
Determine the Thevenin equivalent voltage V_{Th} and resistance R_{Th} w.r.t. terminals $a - b$.	
24	V_{Th}
25	R_{Th}



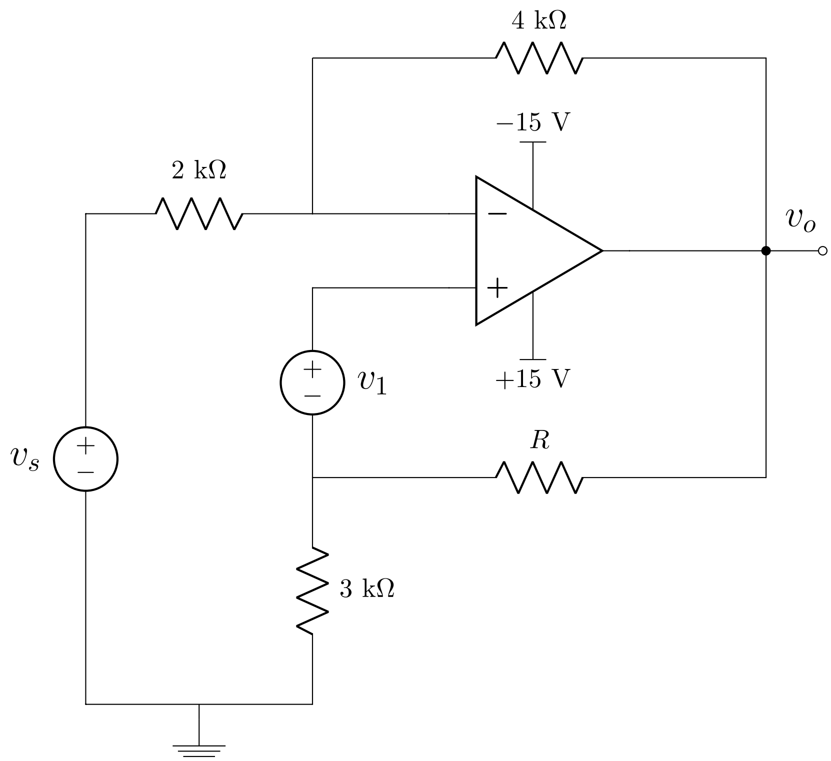
Questions 26–27 [4]	
Determine the Norton equivalent current I_N and resistance R_N w.r.t. terminals $a - b$.	
26	I_N
27	R_N



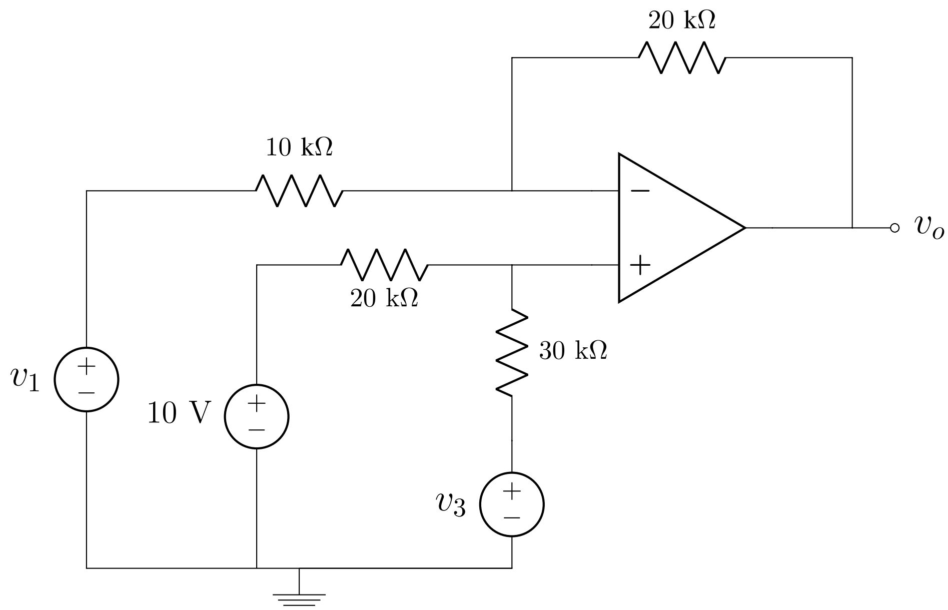
Question 28 [2]	
28	A Norton equivalent circuit is connected to a load resistor R_L, and the maximum power P_{max} is delivered to the load. If the voltage across R_L is 25 V and $P_{max} = 800$ mW, find the Norton equivalent current I_N.

SECTION 2 [46]

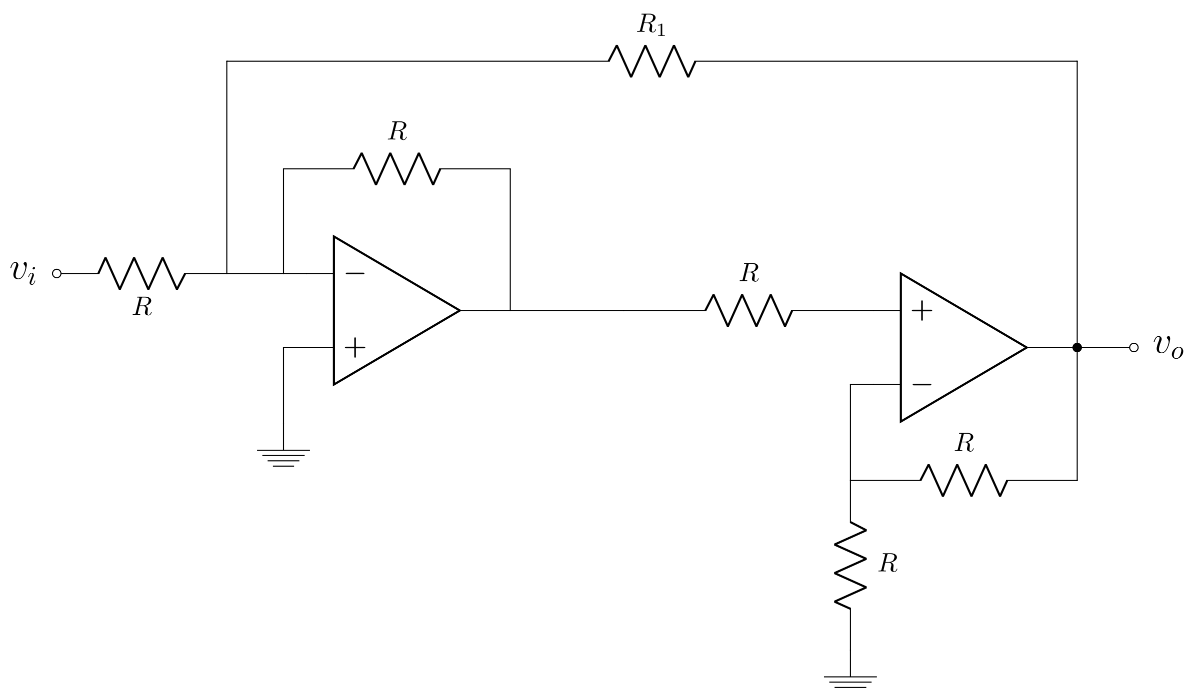
Questions 29–30 [4]	
29	Find v_s if $v_1 = 2$ V, $R = 5$ kΩ, and $v_o = -4$ V.
30	If $v_1 = 0$ and $R \to \infty$, find the maximum value of v_s before the op amp enters saturation.



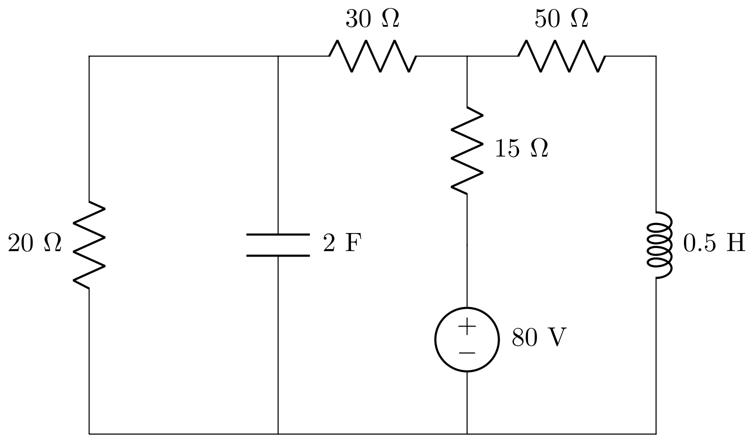
Questions 31–32 [4]	
31	Find v_o if $v_1 = 0$ V and $v_3 = 4$ V.
32	Find v_1 if $v_o = 10$ V and $v_3 = 0$ V.



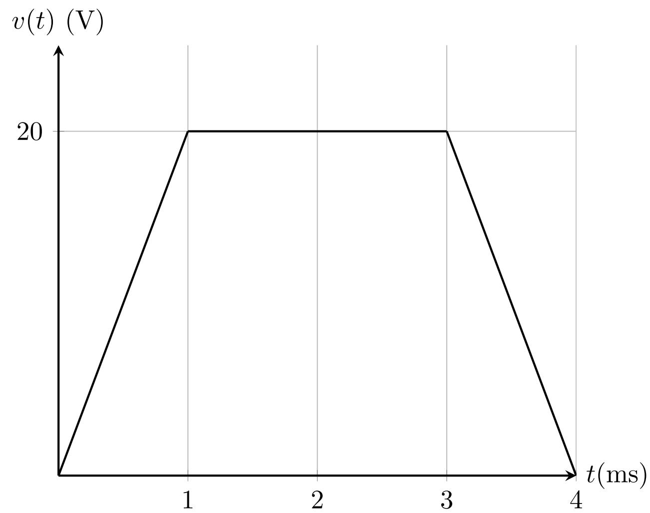
Questions 33–34 [4]	
33	If $R_1 \to \infty$, then find $\frac{v_o}{v_i}$.
34	If $R_1 = R$ and $v_o = 10$ V, then find v_i.



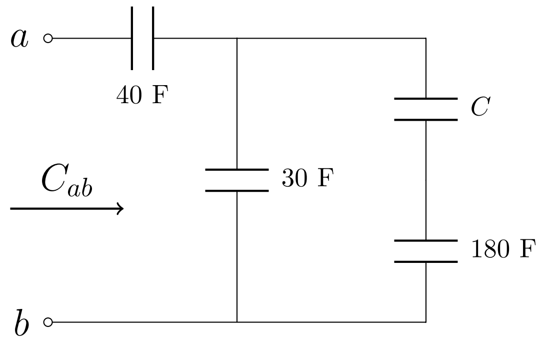
Questions 35–36 [4]	
The circuit has reached a steady state.	
35	Find the energy stored in the capacitor (E_C).
36	Find the energy stored in the inductor (E_L).



Questions 37–38 [2]	
The voltage across a $2\ \mathrm{mF}$ capacitor is shown below. Determine the current through the capacitor at $t = 0.5\ \mathrm{ms}$ and $t = 2\ \mathrm{ms}$.	
37	$i(0.5\ \mathrm{ms})$
38	$i(2\ \mathrm{ms})$



Question 39 [2]	
39	Find C if the equivalent capacitance w.r.t terminals $a - b$ (C_{ab}) is 30 F.



Questions 40–42 [4]		
Given two sinusoid signals $$v_1(t) = 5\cos\left(2t - 53^\circ\right),$$ $$v_2(t) = 12\cos\left(2t - 30^\circ\right).$$		
40	Choose the correct statement. 1) v_1 leads v_2. 2) v_2 leads v_1. Write 1 if statement 1) is correct **or** 2 if statement 2) is correct.	[1]
41	Find the phase angle θ between v_1 and v_2 (with $\theta > 0$).	[1]
42	Use phasors to find $\bar{V}_T = \bar{V}_1 - \bar{V}_2$ in **polar** form.	[2]

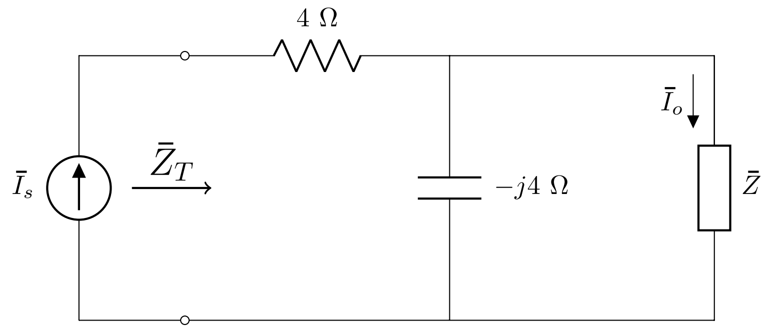
Questions 43–44 [4]	
43	Simplify the following expression and give the answer in **rectangular** form (* refers to conjugation).

$$\bar{V} = \left(\frac{10 + j20}{3 + j4}\right)^2 \sqrt{(10 + j5)^* \, (16 - j20)}$$

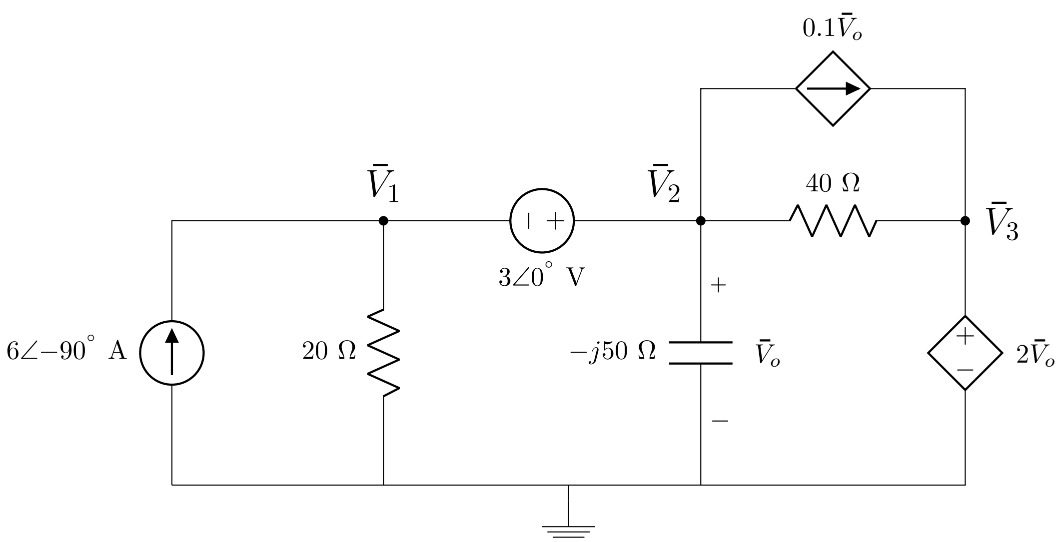
44	Find $\bar{I}_2$ in **polar** form.

$$\begin{bmatrix} 20\angle{-30^\circ} & 2 - j3 \\ -5 & 3\angle 45^\circ \end{bmatrix} \begin{bmatrix} \bar{I}_1 \\ \bar{I}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -5\angle 0^\circ \end{bmatrix}$$

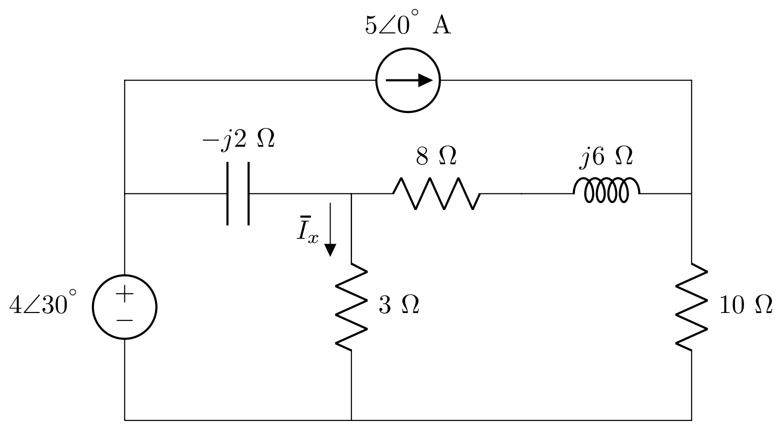
Questions 45–46 [4]	
45	Find $\bar{Z}$ in **rectangular** form if $\bar{Z}_T = 20 + j12\ \Omega$.
46	If $\bar{I}_s = 4\angle 90^\circ$ A and $\bar{Z} = 4 - j4\ \Omega$, find $\bar{I}_o$ in **rectangular** form.



Questions 47–50 [4]			
Use nodal analysis to set up two equations of the following form: $$k\bar{V}_1 + l\bar{V}_2 = -3,$$ $$m\bar{V}_1 + (-4 + j15)\bar{V}_2 = n.$$ Find the complex values k, l, m, and n in **rectangular** form.			
47	*k*	**48**	*l*
49	*m*	**50**	*n*

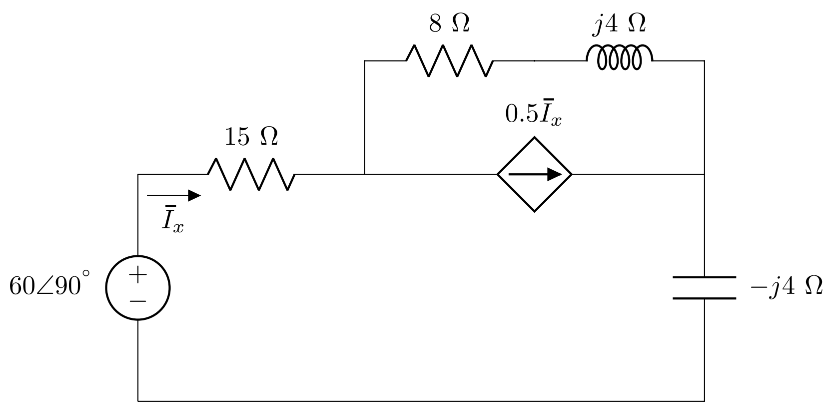


Question 51 [2]	
51	Find $\bar{I}_x^V$, the contribution of the voltage source to $\bar{I}_x$, in **rectangular** form.

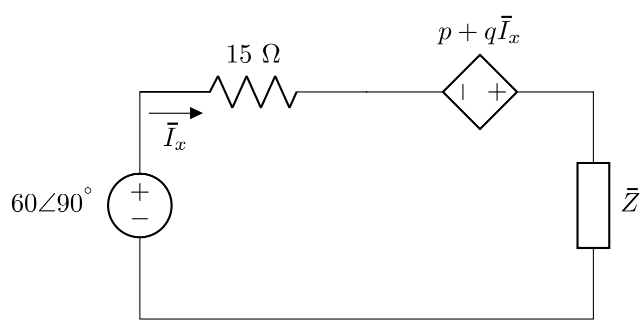


Questions 52–54 [4]		
Use source transformation to determine p, q, and $\bar{Z}$ in ***circuit B*** (in **rectangular** form) such that it is equivalent to ***circuit A***.		
52	p	[1]
53	q	[1]
54	$\bar{Z}$	[2]

Circuit A:



Circuit B:



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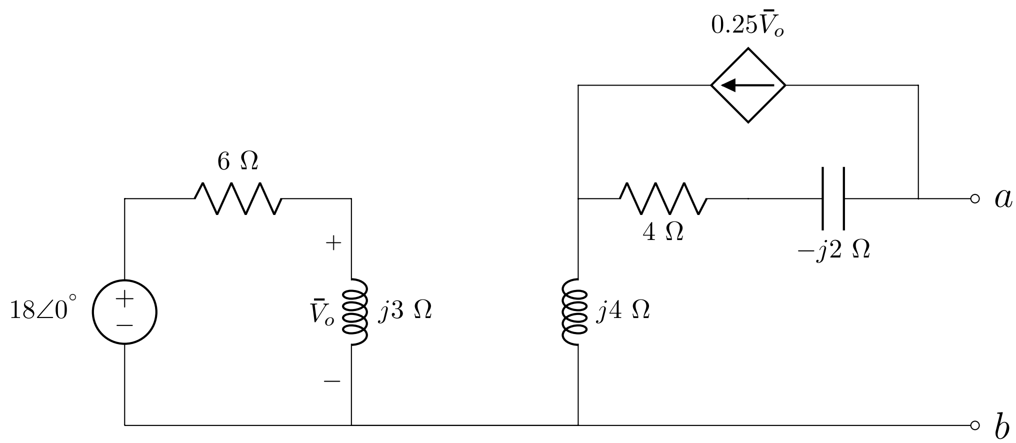
Questions 55–56 [4]	
Find the Thevenin equivalent voltage $\bar{V}_{Th}$ and Norton equivalent current $\bar{I}_N$ w.r.t. terminals $a - b$ in **rectangular** form.	
55	$\bar{V}_{Th}$
56	$\bar{I}_N$



PRACTICE PAGE

Assessment ID	2	0	1	9	E	B	N	1	1	1	E	0	1	

Round off answers to 3 significant digits when applicable.

*All angles must be given in degrees in the range $[-180^\circ, 180^\circ]$. For complex numbers in polar form, unit of angle $(^\circ)$ must be given in the **Answer/**numeric field. For **all other cases**, the unit of angle $(^\circ)$ must be given in the **Unit/**text field.*

	Answer	Unit	Marks
SECTION 1			
01	$E =$		2
02	$Cost =$		2
03	$P_I =$		2
04	$P_V =$		2
05	$R_{ab} =$		2
06	$R_{cd} =$		2
07	$R_2 =$		2
08	$a =$	*No units required*	1
09	$b =$		1
10	$c =$		1
11	$d =$		1
12	$e =$		1
13	$f =$		1
14	$g =$		1
15	$h =$		1
16	$i =$		1
17	$j =$		1

	Answer	Unit	Marks
18	$I_1 =$		2
19	$V_{CE} =$		2
20	$V_x^V =$		2
21	$V_x^I =$		2
22	$V_1 =$		2
23	$R_1 =$		2
24	$V_{Th} =$		2
25	$R_{Th} =$		2
26	$I_N =$		2
27	$R =$		2
28	$I_N =$		2
SECTION 2			
29	$v_s =$		2
30	$v_s =$		2
31	$v_o =$		2
32	$v_1 =$		2
33	$\frac{v_o}{v_1} =$	*No units required*	2
34	$v_i =$		2
35	$E_C =$		2
36	$E_L =$		2
37	$i(0.5 \text{ ms}) =$		1
38	$i(2 \text{ ms}) =$		1
39	$C =$		2

	Answer	Unit	Marks
40	$Correct\ Statement =$	*No units required*	1
41	$\theta =$		1
42	$\bar{V}_T =$	*No units required*	2
43	$\bar{V} =$		2
44	$\bar{I}_2 =$		2
45	$\bar{Z} =$		2
46	$\bar{I}_o =$		2
47	$k =$	*No units required*	1
48	$l =$		1
49	$m =$		1
50	$n =$		1
51	$\bar{I}_x^V =$		2
52	$p =$	*No units required*	1
53	$q =$		1
54	$\bar{Z}_T =$		2
55	$\bar{V}_{Th} =$		2
56	$\bar{I}_N =$		2